

Understanding the mechanism of ageing and a method to improve the ageing resistance of conducting PEDOT:PSS films

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ARTICLE INFO

Article history:

Received 4 July 2019

Received in revised form

30 September 2019

Accepted 12 November 2019

Available online 14 November 2019

Keywords:

PEDOT:PSS

Conducting polymer

Accelerated ageing

Life-time

Shearing

Polaronic states

ABSTRACT

Accelerated thermal ageing studies were carried out on poly(3,4-ethylenedioxythiophene): poly(styrene sulfonate)(PEDOT:PSS) thin films upto 150 °C to predict its life-time. The effect of thermal ageing on the chemical structure and the electronic structure is investigated. Changes in the conductivity of PEDOT:PSS films with ageing are studied along with UV–Vis, ATR-FTIR, Raman spectroscopy and AFM analysis. Conductivity at different temperatures shows an exponential decrease with time in the case of PEDOT:PSS (1.6PSS). Based on this data, the life-time of PEDOT:1.6PSS thin films at ambient conditions is predicted. The analysis of changes in the chemical structure using FTIR spectroscopy and the corresponding changes in the electronic structure using UV–Vis data shows changes in the polaronic states during ageing. Corresponding changes in the Raman spectroscopy data show oxidized-neutral transitions occurring during ageing which results in the conductivity changes. It is observed that unidirectional shearing of PEDOT:1.6PSS dispersion during film formation improved the thermal ageing characteristics. The ageing resistance of these films became comparable to that of PEDOT:2.5PSS films with higher PSS ratio. AFM studies show the effect of ageing on the surface morphology of films and shearing of films during formation can result in better ageing resistance and stable morphology.

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1. Introduction

Among the conjugated polymers, PEDOT:PSS is a highly promising material for several applications in the form of thin films [1]. Applications include electroluminescent devices, light emitting diodes [2,3], polymer solar cells [3–9], perovskite solar cells [10], tandem solar cells [11], electro-chromic windows [12], sensors (pressure, stress, and humidity) [13], actuators [14], thermoelectric devices [15], antistatic coatings, transistors [16] etc. In opto-electronic devices, PEDOT:PSS is used as a hole injection layer and as an electrode. PEDOT:PSS is shown to have potential to replace ITO based anode in devices, with different additives and post treatment of thin films [17,18]. ITO replacement not only reduces the cost of the device but also makes the device more flexible and robust. However, the use of PEDOT:PSS in multi-layered opto-electronic devices is found to reduce their life time [19]. Hence, to improve the life of the device, it is necessary to understand the degradation of the material under end-use application conditions.

Thermal ageing in air at different temperatures for varying durations is one such method. Thermal stability studies carried out so far on PEDOT:PSS films can be classified into two categories: (1) thermal treatment of already formed films to high temperature baking in air/N₂ atmosphere to understand its effect on the electrical/electronic properties of the material (which cannot be classified as ageing) and (2) accelerated thermal ageing studies in air for different durations to understand the contribution of degradation to electrical/electronic properties. Although there are some differences in the approach, both these approaches have shown significant effect on the electronic conductivity of the material. PEDOT:2.5PSS (PH1000) films showed increase in surface resistance when aged at 85 °C and 85% RH. However, under similar conditions, when ethylene glycol is added as secondary dopant (5% EG) PEDOT:2.5PSS films were shown to be stable upto 1000 h [17]. Recently, Stepien et al. [20] have studied the thermal operating window for PEDOT:2.5PSS by subjecting the films to thermal degradation over a wide range of temperatures upto 190 °C. The thermal degradation of PEDOT:PSS was shown to be a complex process involving morphological and chemical changes which could be captured using an exponential decay function that needed to be modified for temperatures >163 °C. These films had 5 wt%

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DMSO as secondary additive.

While, these accelerated thermal ageing studies were carried out in hot air oven on already prepared films, thermal degradation during annealing (thermal treatment) of films during processing has also been analyzed. The effect of non-ionic surfactant (Triton-X: TX) in PEDOT:PSS (PH1000) was explored during annealing (60 °C and 120 °C) on a hot plate [21]. This showed that TX weakens the ionic interaction, leading to the phase separation of PEDOT and PSS. The remaining TX in the film acts as a plasticizing agent, increasing the lamellar packing of alternate PEDOT and PSS upon annealing under ambient conditions, which consecutively enhances the life-time of the material [21]. The effect of rate of heating from 80 to 450 K on PEDOT:PSS (PH500) is also reported at two different heating rates. Heating for 60 min at 5 K/min allows the crystallization to develop more efficiently than heating at 15 K/min, which helps in improving the ageing resistance [22]. The effect of annealing (100 °C–250 °C for 20 h) was studied on PEDOT:PSS (Al4083,1:6) films in air and in nitrogen atmosphere. This showed that conductivity decreased faster for samples annealed in air atmosphere compared to nitrogen atmosphere. This was attributed primarily to the water uptake from air due to the hygroscopic nature of the PSS in the films [23]. Though there are various reported works on the thermal ageing and the effect of thermal treatment on PEDOT:PSS thin films, a systematic study to understand the chemical and electronic structural changes due to ageing is lacking. Accelerated thermal ageing is widely used to predict the life-time of polymers in various applications and to understand the physico-chemical changes during ageing. UV–Vis, ATR–FTIR, Raman spectroscopy along with conductivity studies are used in the present work to understand the kinetics of ageing and to predict the life-time of PEDOT:PSS thin films. Changes in the chemical and electronic structure of PEDOT:PSS thin films due to thermal ageing is used for explaining the changes in conductivity. The effect of various PSS ratios (1.6PSS and 2.5PSS), effect of secondary additives such as, dimethyl sulfoxide (DMSO) and ethylene glycol (EG), effect of humidity and the effect of application of a unidirectional shear during the film formation on the ageing behavior are also investigated in the present work. AFM measurements before and after ageing are used to investigate the morphological changes in the films due to ageing. These results are expected to help in the prediction of life-times as well as to find the appropriate additives/processing methods to improve the thermal stability of the PEDOT:PSS films.

2. Experimental

Two grades of commercially available PEDOT:PSS are used in the study. One with 1:1.6 ratio of PEDOT to PSS (1.3 wt % solids; dispersion in water) represented as PEDOT:1.6PSS and another with 1:2.5 ratio of PEDOT to PSS (1 wt % solids; dispersion in water) represented as PEDOT:2.5PSS. Both were purchased from Sigma Aldrich (Bengaluru, India). Dimethyl sulfoxide (DMSO), 99% purity was purchased from Merck (Mumbai, India) and ethylene glycol (EG), 98% purity was purchased from Rankem (RFCL, New Delhi, India). The PEDOT:PSS dispersions were stirred at room temperature (30 ± 2 °C) and sonicated for 20 min. Secondary additives such as DMSO and EG were added to this dispersion at 20 vol/vol %. A set of glass slides (1×1 cm² and 2×2 cm² size) were cleaned using soap solution and acetone in ultra-sonication bath for 15 min each, and washed with deionized (DI) water in between the soap and acetone cleaning. These slides were then fine cleaned using piranha solution, followed by washing with DI water 2–3 times to clean all acid residues and finally cleaned using isopropyl alcohol (IPA) in ultra-sonication bath. To make them ready for use, these glass slides were then dried on a hot plate. PEDOT:PSS thin films were

spin coated on the clean glass at 1000 rpm for 60 s followed by annealing on the hot plate at 120 °C for 10 min. These samples were then stored in a desiccator.

Thermal ageing of the PEDOT:PSS thin films were carried out at 80 °C, 100 °C, 120 °C and 150 °C in hot air oven (Mettler GmbH + Co. Germany). Surface resistance was measured using four probe conductivity method (Keithley 6221 current source and 2182A nanovoltmeter) before and after ageing. Four silver probes (1 mm diameter) were used. For better contact, a fixed pressure was applied on the probes. These silver probes were kept 2 mm apart, with the middle two to measure voltage, and the outer two to supply current [24]. Chemical changes after ageing were studied using UV–Vis (Jasco UV-Vis spectrometer, wavelength range: 400–1100 nm), ATR–FTIR spectroscopy (Bruker Alpha, wave number range: 2000–700 cm^{−1}) and Raman spectroscopy (Bruker RFS 27; with wave number range 950 cm^{−1} to 1800 cm^{−1}) analysis on spin coated thin films aged for different durations. Atomic force microscopy (AFM–SPM, Asylum Research, Oxford Instruments, USA.) was used to characterize the morphology of the films before and after ageing. Samples were characterized in tapping mode with AC240TM–R3 probe (silicon lever with Al coating and silicon tip of 28 ± 10 nm radius with Ti/pt: 5/20 coating). All characterizations were carried out after cooling the films to 25 ± 2 °C. UV–Vis, FTIR and Raman spectra were probed at 3–4 different locations on a sample and were repeated on three different samples to arrive at consistent results. Differential scanning calorimetry (DSC from 0 °C to 450 °C with ramp 10 °C/min) and thermo-gravimetric analysis (TGA from 25 °C to 800 °C with ramp of 5 °C/min) (TA instruments) were carried out for the thermal analysis of the films. To shear the dispersion uni-directionally a tensile testing machine, (BISSE Bengaluru, India) with movable arms was used as shown in Fig. 1(a). To shear the films while forming, a fluid drop was placed between two glass slides fixed on two opposite arms of the machine and pulled in opposite direction at 1 mm/s speed (Fig. 1(b)). A double sided adhesive tape was used to keep the glass slides fixed on the moving arms of the instrument during shearing.

0.2 ml of the PEDOT:1.6PSS dispersion was deposited on a clean glass slide and the second clean glass slide was placed above the drop. The top and bottom glass slides were pulled uni-directionally at a rate of 1 mm/s until the drop was spread between the glass slides on an overlapping area marked in Fig. 1(a). It has been shown that unidirectional stretching in the wet condition can result in increased conductivity in PEDOT:PSS [25]. Once the drop was spread by pulling glass slides, the thin film of PEDOT:1.6PSS dispersion annealed at 120 °C for 10 min and stored in a desiccator for 24 h before further experiments. Ageing studies on all these samples were conducted in a hot air oven at 120 °C for 50 h. Surface resistance and UV–Vis spectroscopic measurements were carried out before and after ageing for different durations to understand the kinetics of the degradation process. Further, to understand the effect of humidity on ageing behavior, PEDOT:1.6 PSS films were aged at 85% RH and 85 °C in a humidity chamber (using Climate Chamber, Mettler GmbH + Co. KG, Germany) for different durations. Prior to the resistance measurements, the films were annealed for 10 min at 120 °C to remove any adsorbed moisture from the surface.

3. Results and discussion

Conducting polymers used in opto-electronic devices have reduced life-time and poor stability under standard operating conditions due to their inherent conjugated chemical structure [26]. Accelerated thermal ageing [27] can be used to predict the life-time of such materials by subjecting them to higher temperatures and humidity conditions for varying durations. PEDOT:1.6PSS films

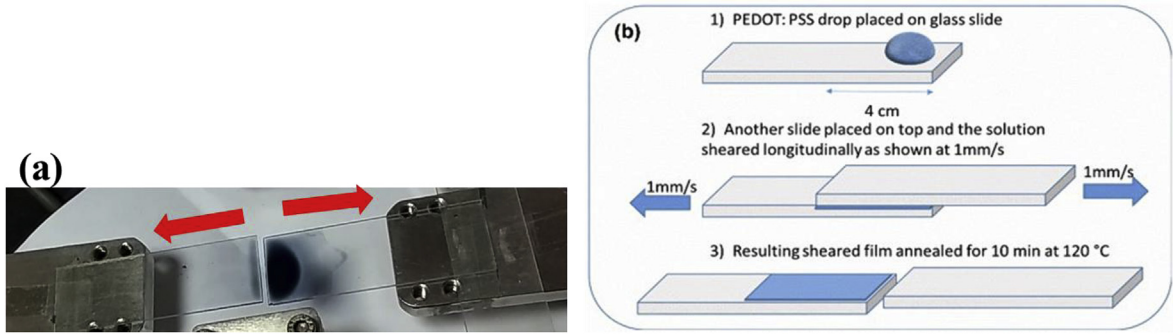


Fig. 1. (a) Experimental set-up for uni-directional shearing (b) Schematic showing shearing of the glass slides with PEDOT:1.6PSS drop placed between the overlapping area of the two slides.

coated on glass substrates were thus subjected to accelerated thermal ageing at different temperatures and for varying durations. The changes in electrical conductivity and chemical structure before and after ageing were studied to arrive at the kinetics and the mechanism of degradation. Conductivity of the PEDOT:1.6PSS films were estimated from the measured sheet resistance using 4-probe conductivity method as follows,

$$\sigma_0 = l/R_0A; \quad \sigma_a = l/R_aA; \quad \sigma_N = \sigma_a/\sigma_0 = R_0/R_a \quad (1)$$

where, σ_0 is the conductivity before ageing, σ_a is the conductivity after ageing, σ_N is the normalized conductivity, and R_0 and R_a are the corresponding sheet resistance before and after ageing. l is the distance between the electrodes (constant during the measurements), and A is the cross sectional area of the sample which is the product of the film thickness (a negligible variation is observed after ageing (Fig. S3) and hence assumed to be constant) and probe contact length l (constant for each sample). Hence with constant A and l , the ratio of conductivities is equivalent to the ratio of resistances as shown in Equation (1).

Change in conductivities for PEDOT:1.6PSS thin films thermally aged at 80 °C, 100 °C, 120 °C and 150 °C are shown in Fig. 2. From these results, the kinetics of the degradation process is arrived at and the accelerated life time (ALT) is predicted [27–29].

An empirical model is used to determine the degradation constants. Accordingly, the conductivity after ageing (σ_a) can be related to the conductivity before ageing (σ_0) through an exponential function as,

$$\sigma_a = \sigma_0 \exp(-k_d t) \quad (2)$$

where, k_d is the degradation constant. The obtained degradation constants for PEDOT:1.6PSS at different temperatures are given in Table 1. An Arrhenius type of equation can be used to describe the experimentally observed degradation process [29] which is as follows:

$$k_d = A \exp\left(-E_a/k_B T\right) \quad (3)$$

Where, E_a is the activation energy required for the degradation, k_B is the Boltzmann constant, T is the temperature and A is a constant dependent on the degradation mechanism and the experimental conditions [28,29]. Typical processes described by the Arrhenius model for the reduction in conductivity are material related, such as, the trapping of electrical carriers as well as structural and chemical changes in the polymers [28]. From the degradation constants determined at two different temperatures (test temperature T_a and a reference temperature T_0), the acceleration factor k_{Acc} , and hence the life-time can be determined as follows,

$$k_{Acc} = \frac{k_d(T_a)}{k_d(T_0)} = \exp\left[\frac{E_a}{k_B} \left(\frac{1}{T_0} - \frac{1}{T_a}\right)\right] = \frac{L1}{L2} \quad (4)$$

where, $L1$ is the life-time (h) at the reference temperature T_0 and $L2$ is life-time (h) at the test temperature T_a . $L1$ is also defined as the reciprocal of $k_d(T_0)$ and $L2$ is the reciprocal of $k_d(T_a)$. From Table 1, it can be observed that the degradation constant, k_d increases with increasing temperature indicating increasing degradation rates. From the plot of k_{Acc} vs $(1/T_0 - 1/T_a)$ shown in Fig. 3(a), the activation energy for the degradation process is estimated as, 0.4 eV. The life-time showed a reduction of approximately three orders of magnitude after ageing at 150 °C (Fig. 3(b)).

Since most of the devices are used at ambient temperature conditions, it is worthwhile predicting the degradation behavior under such conditions. Hence the degradation constant for PEDOT:1.6PSS thin films is estimated for 25 °C using the values of

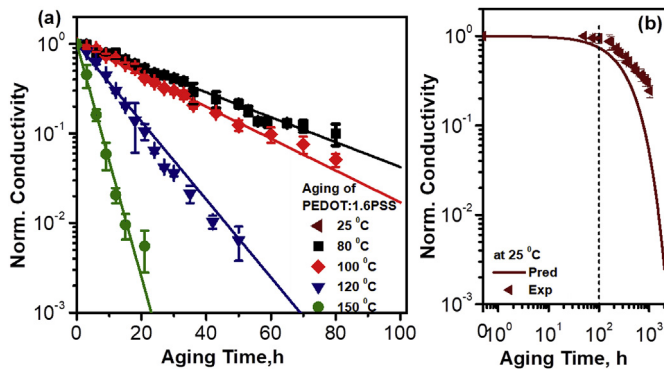


Fig. 2. (a) Change in conductivity of PEDOT:1.6PSS films with thermal ageing at different temperatures. Continuous solid lines represent empirical model based on equation (2); (b) predicted and experimental normalized conductivity at 25 °C.

Table 1
Degradation constant (k_d) and acceleration factor (k_{Acc}) for PEDOT:1.6PSS films.

Temp in °C	k_d (1/sec) ($\times 10^{-5}$)	k_{Acc} at $T_0 = 80$ °C	k_{Acc} at $T_0 = 25$ °C
25	0.01	—	1
80	0.87	1	87
100	1.06	1.22	106
120	2.78	3.20	278
150	8.81	10.12	881

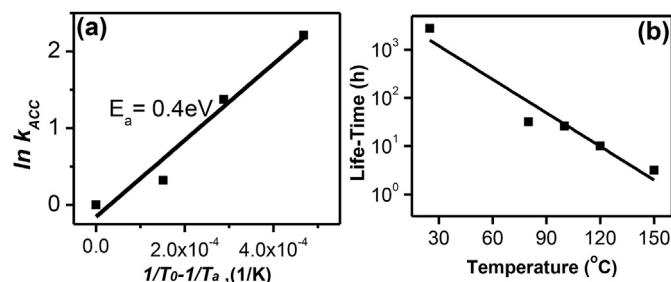


Fig. 3. (a) Variation of k_{Acc} with temperature (at reference temperature $T_0 = 80^{\circ}\text{C}$) and (b) Life-time variation with temperature.

k_{Acc} obtained at different temperatures (Table 1). From this data, the change in the conductivity with time at 25°C is predicted and is shown in Fig. 2(b). This shows that the PEDOT:1.6PSS films can operate without significant change in conductivity for more than 200 h at 25°C . To validate this prediction, conductivity changes were experimentally determined for 1000 h at $25 \pm 3^{\circ}\text{C}$ (films stored in a desiccator) and compared with the prediction in Fig. 2(b). The predicted values are comparable to the experimentally determined conductivity. In the case of PEDOT:2.5PSS (PH1000), it was shown that σ_a/σ_0 reduces to 0.5 within one day (40% RH and 22.5°C) [30], whereas, for PEDOT:1.6PSS there was not much variation (0.94) after 100 h at 25°C (stored in a desiccator).

The contributions from the charged species to the reduction in conductivity with increase in temperature in PEDOT:1.6PSS were studied using UV–Vis absorption spectroscopy before and after ageing (Fig. 4 (a)–(c)). Unaged PEDOT:1.6PSS sample shows characteristic absorptions at 600 nm, 910 nm and 1250 nm (and above), indicative of the neutral (non-ionic), polaron (cation) and bipolaron (di-cation) states in PEDOT respectively [31]. However, as can be observed from Fig. 4(a), (b) and (c), the intensity of these absorptions changes as ageing progresses in PEDOT:1.6PSS at temperatures, 120°C , 150°C and 200°C . Ageing at 120°C shows slow formation of polaron state and insignificant neutral state (Fig. 4(a)), whereas, ageing carried out at 150°C shows fast formation of polaron and slow change in the neutral state (Fig. 4(b)). At 200°C , the experimental evidence indicates a very quick progression to the neutral state with the polaron state existing only for a very short time ($<12 \text{ h}$) (Fig. 4(c)). The changes in the polaron and neutral states with ageing time at 120°C is summarized in Fig. 5 (a). It can be observed that the higher rates of drop in conductivity at higher temperatures (120°C and 150°C) than at lower temperature (80°C) can be due to the formation of neutral states of PEDOT at higher temperatures. Since the polaronic states almost vanishes after few

hours of ageing at 200°C (Fig. 4 (c)), it shows the material must have undergone significant chemical degradation at that temperature. To understand this better, DSC and TGA studies were carried out on the PEDOT:1.6PSS films and is shown in Fig. 6(a) and Fig. 6(b) respectively.

The sharp peak observed in the DSC around 120°C indicates the presence of volatile moisture during the first heating cycle and a 20 wt% (approx) loss in the TGA analysis [32–35]. Peaks observed between 200°C and 400°C in DSC and the weight loss observed in the TGA indicate PEDOT decomposition. The peaks observed above 400°C in DSC signifies the decomposition of PSS, as reported earlier [36,37].

To understand the chemical changes associated with the decrease in conductivity during ageing at 120°C , ATR-FTIR studies were carried out. Fig. 7(a) and (b) show the FTIR spectra of the PEDOT:1.6PSS films before and after thermal ageing at 120°C for 50 h. The bands at 860 cm^{-1} , 727 cm^{-1} , 697 cm^{-1} , 830 cm^{-1} and $930\text{--}943 \text{ cm}^{-1}$ corresponding to the --C--S-- stretch in PEDOT, remained unchanged throughout the ageing period of 50 h. This shows that the --C--S-- of thiophene ring of the PEDOT is not affected by the thermal ageing at 120°C for 50 h. For further FTIR analysis, spectra were divided into distinct regions to identify the significant changes during ageing. Significant changes can be observed in region $1800\text{--}700 \text{ cm}^{-1}$ (Fig. 7(a)) and the highlighted region between 1260 cm^{-1} to 1550 cm^{-1} (Fig. 7(b)). In Fig. 7(a), bands at 1047 and 1183 cm^{-1} are due to --C--O--C-- stretch [38]. These two disappear after 6 h and 18 h ageing respectively. However a narrow band around 1190 cm^{-1} was observed after 18 h, which can be attributed to the presence of oxyethylene after 50 h of ageing. The characteristic absorption corresponding to the thiophene ring in PEDOT, C--C at 1290 cm^{-1} and C=C at 1520 cm^{-1} bands indicate the oxidized state. These two bands shift to 1323 cm^{-1} and 1484 cm^{-1} respectively after 50 h of ageing at 120°C indicating reduced state [38]. However, the bands at $1600\text{--}1660 \text{ cm}^{-1}$ remained unchanged indicating the benzene ring in the PSS chains remaining intact. The significant changes observed in the FTIR spectra are due to the changes in the PEDOT chain as summarized in Table S2. The benzene ring ($1600\text{--}1660 \text{ cm}^{-1}$) and the stretch due to --SO_3 ($1122\text{--}1162 \text{ cm}^{-1}$) corresponding to the PSS remains intact indicating its stability against ageing [36,37].

3.1. Effect of PSS ratio, additives and solution shearing on ageing

To enhance the conductivity in PEDOT:PSS, several approaches have been proposed [1,39]. It has been reported that primary dopant PSS as well as secondary additives, such as high boiling solvents [40–42], and films formed after solution shearing

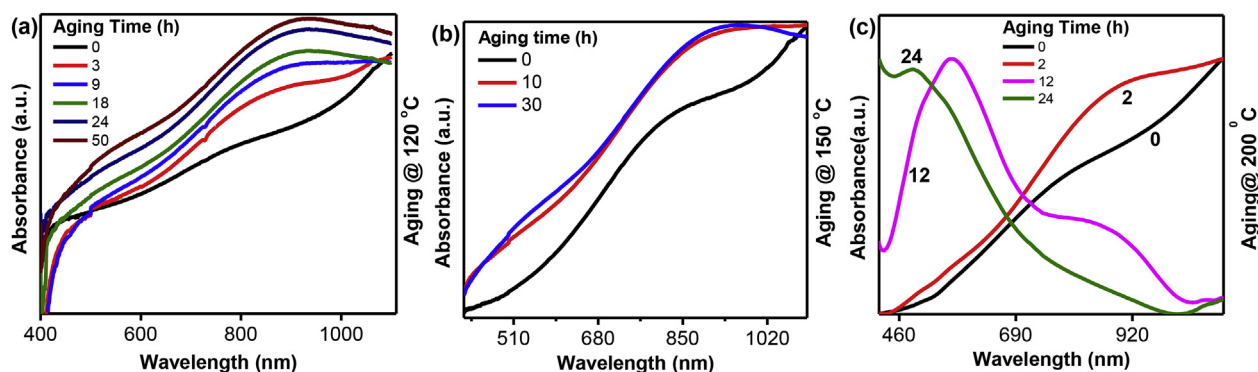


Fig. 4. UV–visible spectra showing changes in the polaron (at 910 nm) and neutral (at 600 nm) states in the PEDOT:1.6PSS thin films before and after ageing (a) at 120°C , (b) at 150°C and (c) 200°C .

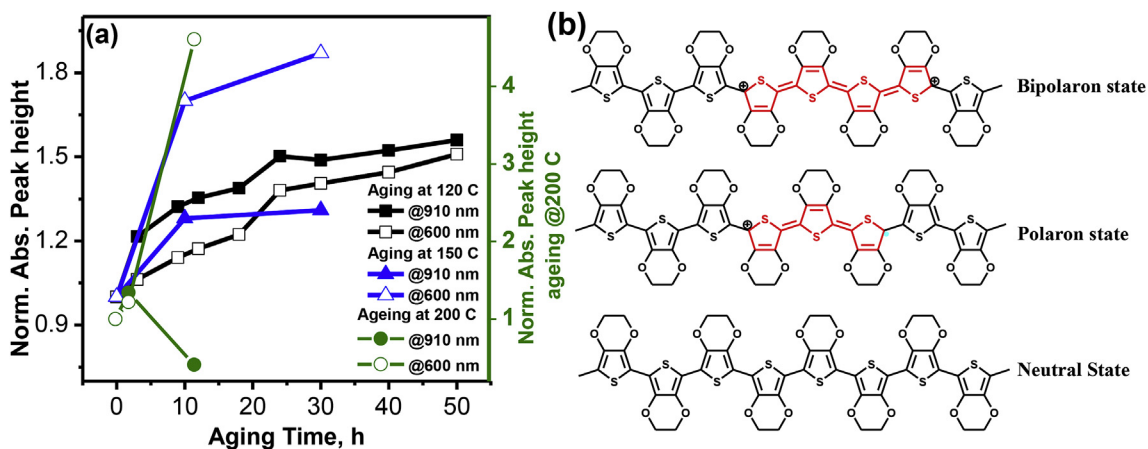


Fig. 5. (a) Changes in UV visible absorption intensity at 910 nm and 600 nm indicating changes in the polaron and neutral states during thermal ageing at 120 °C (b) Schematic of the bipolaron, polaron and the neutral states in PEDOT.

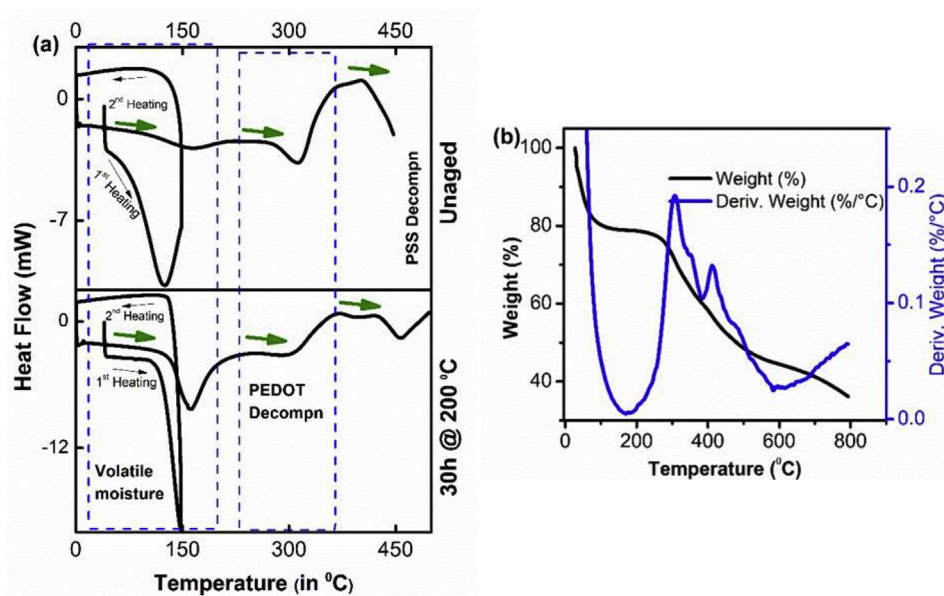


Fig. 6. (a) DSC thermograms of PEDOT:1.6PSS films (unaged and aged at 200 °C for 30 h) (b) TGA of unaged PEDOT:1.6PSS

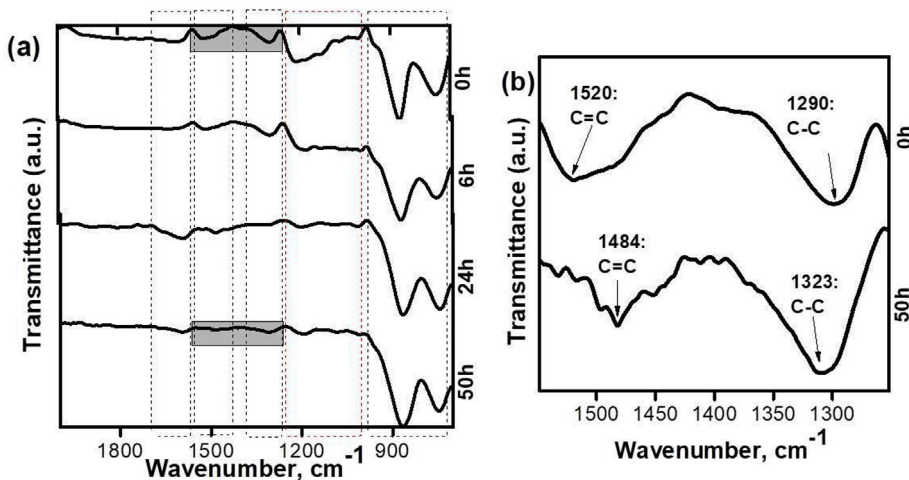


Fig. 7. FTIR spectra of PEDOT:1.6PSS with ageing at 120 °C (a) full spectra indicating significant changes with ageing time. (b) Highlighted region from (a) is magnified to show significant changes observed between 1260 and 1550 cm⁻¹.

followed by post treatment with methanol show significant increase in the conductivity of PEDOT:PSS [25]. To understand the effect of higher PSS content, secondary additives (DMSO and EG), and solution shearing during film formation on ageing characteristics, films were aged at 120 °C and studied.

Fig. 8(a) shows the conductivity variation with ageing and Fig. 8(b) shows the associated changes in the electronic structure observed from the UV–Vis spectra. Absorption peaks at 910 nm and 600 nm correspond to the polaron state and the neutral state respectively. UV–Vis data of unaged films of PEDOT:2.5PSS, PEDOT:1.6PSS with secondary additives (DMSO/EG) and films formed after shearing do not show much difference (Fig. S1) [43,44]. Upon ageing at 120 °C, the PEDOT:2.5PSS shows higher resistance to thermal degradation, in comparison to the films with DMSO and EG (Fig. 8(a)). As can be observed from Fig. 8(b), neutral states found to be less evolving in PEDOT:2.5PSS, whereas, the PEDOT:1.6PSS films (with and without additives) show significant evolution of neutral state. The rate of formation of neutral state is higher in the case of films with secondary additives in comparison to PEDOT:2.5PSS and sheared films after ageing. Surprisingly, there is less evolution of polaron state and neutral states in sheared films than in unaged PEDOT:1.6PSS films. When the formation of polaron states are compared, PEDOT:2.5PSS and PEDOT:1.6PSS show faster formation of polaron state than sheared and DMSO/EG added films. Even though the decrease in conductivity with ageing are similar in PEDOT:1.6PSS films with and without DMSO and EG, the UV–Vis spectra indicate that the formation of polaron and neutral states contribute differently to the conductivity reduction. Theoretical studies suggest that the formation and stability of bipolaron states in conjugated polymers is contributed by different interactions (inter-chain coupling, interactions of charged states, etc.) as well as the configurational entropy [45]. These contributions create the energy difference between the double charged bipolaron and two single charged polarons. For polythiophene, this energy difference is ~0.4 eV which is found to be close to the activation energy obtained from Fig. 3(a). Bipolaron-polaron transitions are possible when the energy difference is low and bipolarons and polarons can also co-exist [45,46]. However, in the present study, the changes in the bipolaron states are not investigated due

to the limitation of the equipment.

Moreover, earlier reports showed that conductive PEDOT grain size decreases due to ageing with a thick PSS insulating barrier between the grains. This increases the hopping distance between PEDOT grains [47]. Fig. 8(a) shows that conductivity decreases exponentially with time in all the PEDOT:PSS samples. However, the rate of reduction in conductivity in the case of higher PSS containing PEDOT:PSS and films of PEDOT:1.6PSS formed after shearing the dispersion are much slower indicating their higher stability against ageing. However, the mechanism seems to be different as can be observed from the UV–Vis absorption data. In the case of PEDOT:2.5PSS and PEDOT:1.6PSS without additives, polaron state increases with ageing.

The ageing kinetics of the PEDOT:PSS films with different PSS ratio, additives and shearing, are compared using the model given in equation (2) and the results are given in Table 2. This shows that the degradation constant is higher when additives such as EG and DMSO are added to PEDOT:1.6PSS. It can also be observed that PEDOT:2.5PSS and dispersion sheared PEDOT:1.6PSS show lowest degradation constants indicating their thermal stability. To understand this interesting observation, further analysis using Raman spectroscopy was carried out.

Raman spectroscopy is one of the most valuable tools for studying the charged states in conjugated polymers. In the present study, it is used to understand the changes in the charged states during thermal ageing of PEDOT:PSS thin films. Fig. 9(a) shows the Raman spectra of unaged and PEDOT:PSS films aged at 120 °C for 50 h. These include 1.6PSS, 2.5PSS, 1.6PSS sheared, 1.6PSS + DMSO, 1.6PSS + EG films. The principal bands for unaged PEDOT:1.6 PSS at:

Table 2

Effect of PSS ratio, additives and solution shearing on the degradation constant (k_d).

PEDOT:PSS with	k_d (1/sec) ($\times 10^{-5}$)
1.6PSS	2.78 ± 0.19
1.6PSS + DMSO	3.75 ± 0.79
1.6PSS + EG	3.93 ± 0.84
1.6PSS + Shear	0.9 ± 0.11
2.5PSS	0.8 ± 0.19

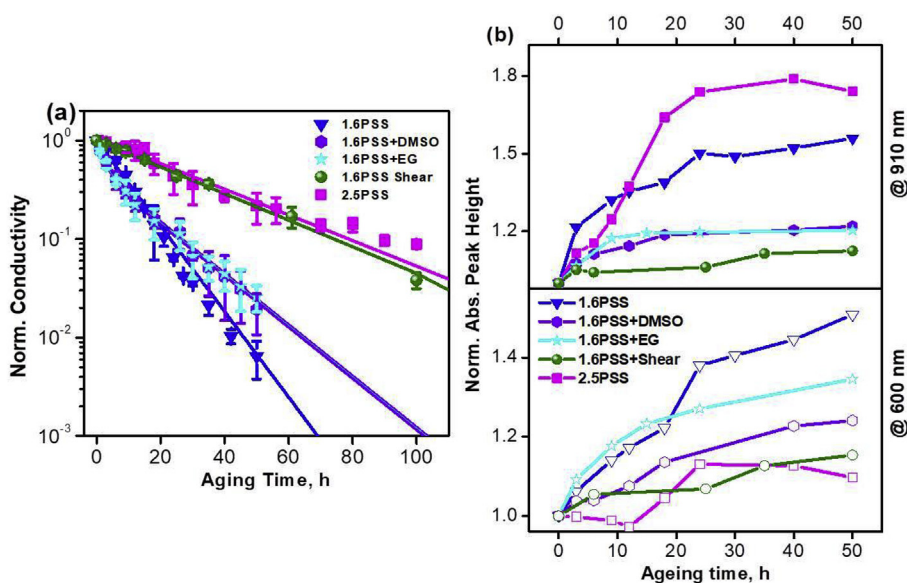


Fig. 8. (a) Effect of ageing on the conductivity of PEDOT: PSS thin films at 120 °C (b) Polaron state (filled symbols) and neutral state (open symbols) concentration data extracted from UV–Vis spectroscopy.

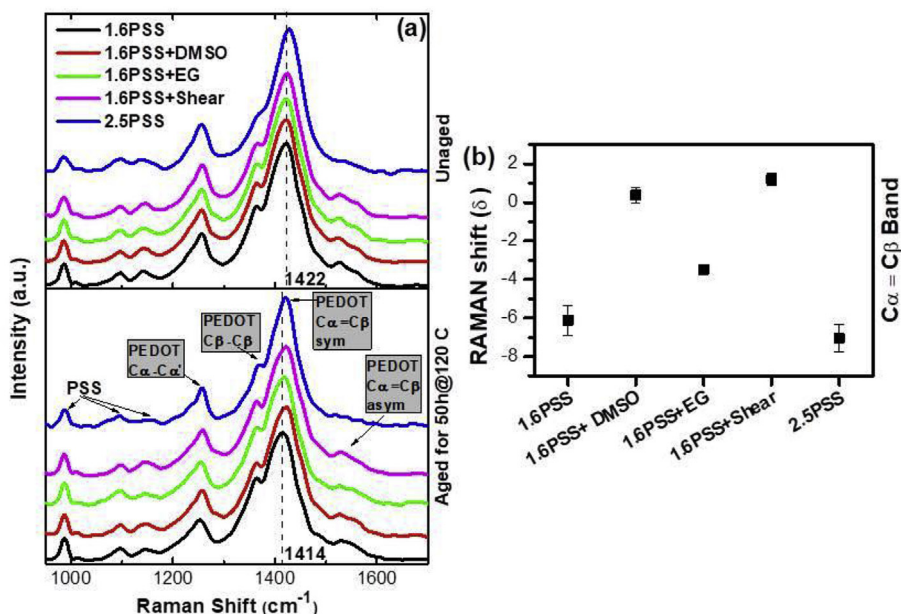


Fig. 9. (a) Raman spectra of PEDOT:1.6PSS thin films, with DMSO, with EG, after shear and PEDOT:2.5PSS; before (top) and after ageing (bottom) at 120 °C for 50 h; (b) Relative shift in Raman spectra at wave number 1422 cm⁻¹ after ageing for 50 h at 120 °C.

1272, 1383, 1452, and 1524 cm⁻¹ are attributed to C_α-C_{α'} inter-ring stretch, C_β-C_β stretching, symmetrical C_α=C_β stretching, asymmetrical C_α=C_β stretching respectively [48–50]. From Fig. 9(a) and Tables S1 and S2 (supporting information), it can be observed that the principal bands for PEDOT:1.6PSS, 1.6PSS + DMSO and 1.6PSS + EG are similar before ageing. However, after ageing, there are shifts in the bands. In the case of 1.6 PSS and 1.6PSS + EG the band at 1257 cm⁻¹ corresponding to C_α-C_{α'} inter-ring stretch, show significant shift after ageing to lower wave number indicating stretch among EDOT monomers. However, such a shift is not observed in the case of aged films of PEDOT:1.6PSS + shear, PEDOT:1.6PSS + DMSO and PEDOT: 2.5PSS films.

The band at 1365 cm⁻¹ corresponding to C_β-C_β stretch for PEDOT:2.5PSS films was not observed before ageing. However after ageing, a significant band could be observed at 1369 cm⁻¹. In the case of PEDOT:1.6PSS, 1.6PSS + DMSO and 1.6PSS + EG and 1.6PSS + shear films, this band was observed before and after ageing without any significant shift. The most important shift is for the symmetrical C_α=C_β stretch at 1422 cm⁻¹ as shown in Fig. 9(b). A shift to lower wave number indicates oxidized (linear structure with C_α-C_β single bond) and shift to higher wave number indicates neutral form (reduced and coiled structure with C_α=C_β double bond) [48,51]. This band shifted slightly from higher to lower wave number for 1.6PSS, 1.6PSS + EG and 2.5PSS, implying the transition from oxidized to neutral structure. Such a shift was not observed in the case of 1.6PSS + DMSO and 1.6PSS + shear samples, implying the presence of stable oxidized state of PEDOT in both these films during ageing. 1527 cm⁻¹ band corresponding C_α=C_β asymmetrical stretch was found to shift to higher wave numbers in all the films except solution sheared films. The reason behind stable oxidized structure in sheared films could be that the charged states in PEDOT are protected by the surrounding thin PSS layer. This could have resulted from the uniform dispersion of PSS in the film due to shearing. It has also been shown that PEDOT and PSS remains electrostatically bound and forms alternate layered/ordered structure after shearing. PSS chains alternately stacked with PEDOT can provide longer life for PEDOT:PSS [16,52]. Hence sheared PEDOT:1.6PSS films show better morphological and thermal

resistance to ageing as observed from this work. However, DMSO/EG addition does not improve the ageing resistance. DMSO/EG addition results in ordered PEDOT domains in PEDOT:PSS. However, since the addition of DMSO and EG, phase separates the PEDOT domains from the PSS [53–57], the PEDOT domains are more vulnerable to degradation during ageing, unlike the sheared films and PEDOT:2.5PSS where the PSS layers prevent the ageing of the PEDOT.

Spin coated films were also characterized using AFM to understand the morphological changes after ageing at 120 °C for 50 h. Fig. 10 (a) shows height, and Fig. 10 (b) shows the phase. Fig. 11 captures the RMS (root mean square) surface roughness in nm and the RMS phase shift in degrees for all the films. The height images show that after ageing, there was a reduction in surface roughness for all the films, as corroborated from the RMS values in Fig. 11. The roughness reduced by a small amount for 1.6PSS (6.1 nm–4.7 nm), sheared (7 nm–5.8 nm) and 2.5PSS (7.5 nm–6.5 nm) films as they were smoother to begin with. However, in the case of DMSO/EG added films, ageing reduced the roughness of the films significantly (10.8–6.5 nm for DMSO and 15.9 to 12.6 nm for EG added films). In the phase images shown in Fig. 10 (b), there are bright and dark areas and the color scale represents lower phase angle to higher phase angle. In polymer systems, higher phase angle represents harder material (brighter areas), and lower phase angle represents relatively softer material (darker areas). In Fig. 10(b), the darker areas indicate the presence of softer PSS shell and the brighter areas correspond to harder PEDOT core. The size of the area indicates the domain size [54]. In the case of PEDOT:1.6PSS, the unaged films show more number of bright areas corresponding to the PEDOT core while the thermally aged films show more number of dark areas corresponding to the PSS shell. Ageing at higher temperatures results in the shrinking of the PEDOT domains. Similarly, in DMSO/EG added films, the unaged films show larger PEDOT domains throughout the films which is known to have more aligned polymer chains. After ageing, the large/ordered grains of PEDOT shrink in size and smaller grains can be found surrounded by larger PSS areas resulting in a reduction of the conductivity. Hence conductivity reduction could be attributed

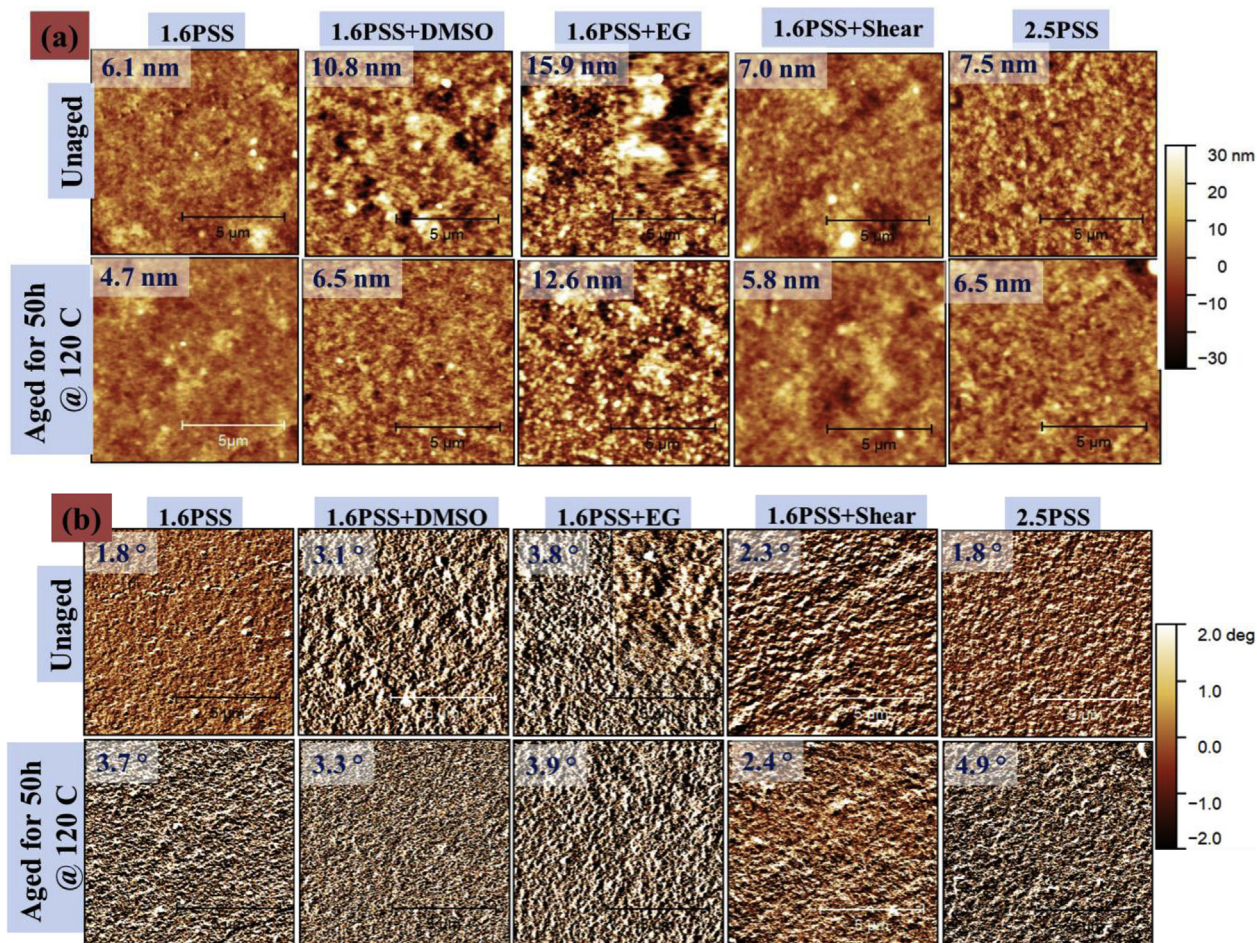


Fig. 10. Atomic force microscopy (AFM) images before and after ageing for PEDOT: 1.6PSS thin films, with DMSO, with EG, after shear and PEDOT: 2.5PSS; (a) height images in nm (b) phase images in degree (scale bar in each window is 5 μm).

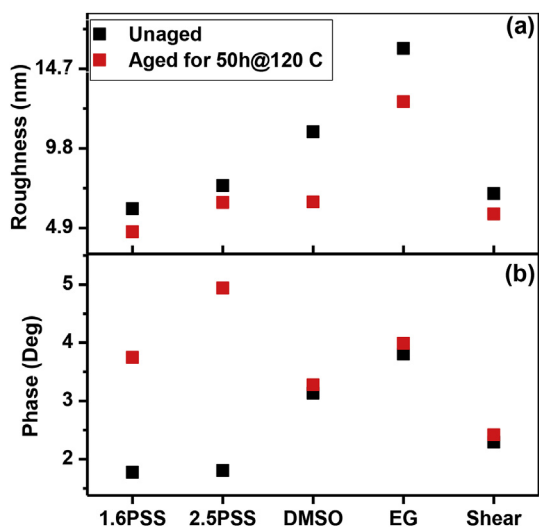


Fig. 11. Morphological parameters obtained from AFM images for PEDOT: 1.6PSS thin films, with DMSO, with EG, with unidirectional shear and for PEDOT: 2.5PSS before and after ageing; (a) surface roughness in nm (b) phase shift in degree.

to the chemical changes in the PEDOT structure as observed from the Raman spectroscopy and FTIR data. The chemical and

corresponding electronic structural changes could be resulting in the domain size change in PEDOT. This can be further corroborated from Fig. 11. Sheared films show the least change in RMS roughness and RMS phase values even after ageing, indicating that these films are morphologically more stable.

Effect of humidity and temperature on conductivity during ageing (85 °C and 85% RH) is shown in Fig. 12. In the presence of humidity, both PEDOT:1.6PSS and PEDOT:1.6PSS + EG show faster deterioration in conductivity. However, PEDOT:1.6PSS + DMSO films, do not show much difference in the reduction in conductivity in the presence of humidity. This shows that the addition of DMSO does not improve the ageing resistance in the presence of moisture. This could be due to the increased presence of PEDOT domains on the surface of the films in DMSO added systems which are shown to be vulnerable to ageing as can be observed from the FTIR and UV–Vis data. In PEDOT:2.5PSS films, though there is an initial increase in the rate of degradation with humidity, the conductivity is not further affected due to this. Uni-directionally sheared films, however, show resistance to ageing in the presence of humidity with significant stability in conductivity values. This could be attributed to the ordered, aligned and uniformly dispersed PSS shell which is thinner due to the shearing action. In the unsheared cases, the thicker PSS domains swell more under humidity, resulting in more separation between the conducting PEDOT domains. This shows that thinner PSS shell could help prevent PEDOT:PSS from ageing faster which can be achieved by mechanical shearing during

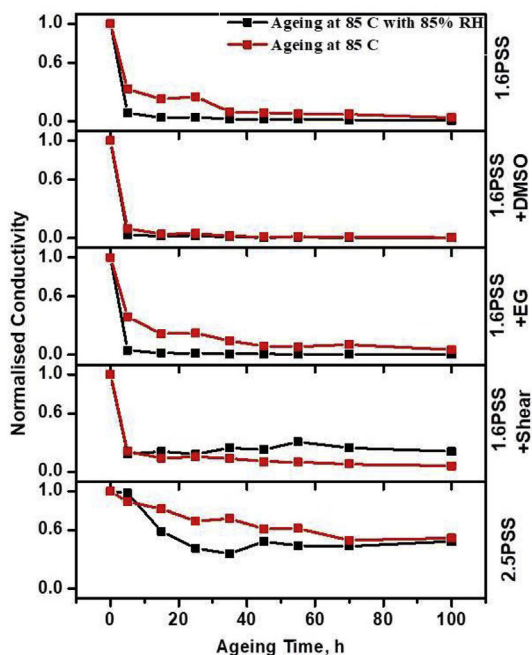


Fig. 12. Effect of humidity during ageing on the conductivity of PEDOT:1.6PSS, PEDOT:1.6PSS + DMSO, PEDOT:1.6PSS + EG, PEDOT:1.6PSS + shear and PEDOT:2.5PSS films. Ageing also compared with ageing at 85 °C in hot air oven.

processing. However, this requires further studies to establish the mechanism.

Schematic shown in Fig. 13 summarizes this morphological changes that contribute to the conductivity changes in PEDOT:PSS thin films during thermal ageing. With this morphological understanding, the higher ageing resistance of PEDOT: 2.5PSS and sheared PEDOT:1.6 PSS films could possibly be attributed to the four differentiating characteristics: a) aggregate size, b) PEDOT chain ordering, c) phase segregation, and d) stable oxidized states in PEDOT.

4. Conclusions

PEDOT:PSS thin films coated on glass substrate were subjected to accelerated thermal ageing at various temperatures to understand the conductivity changes and the associated electronic and structural changes. Conductivity as a function of ageing temperature and time were determined along with UV–Vis, FTIR, Raman spectroscopy and AFM measurements on the aged films. From the accelerated ageing data, life-time predictions could be made at room temperature using a kinetic model, which were validated using experiments. UV–Vis spectra of PEDOT:1.6PSS films during ageing show stable polaronic states formed by the dissociation of bipolarons below 200 °C and more neutral states above 200 °C leading to drastic reduction in conductivity. FTIR Spectroscopy also confirmed the formation of these polaron and neutral states during ageing due to the formation of reduced state of PEDOT during ageing. Thin films of PEDOT:PSS with higher PSS content and sheared films showed better ageing resistance in comparison to films with secondary additives such as, DMSO and EG. From the Raman spectra, 1.6PSS, 1.6PSS + EG, 2.5PSS films show changes from oxidized to neutral structure of PEDOT due to ageing. However, this change was not observed in 1.6PSS + DMSO and solution sheared films. In the case of solution sheared films, this can be attributed as due to the alignment of PEDOT and the uniform distribution of PSS throughout the film. AFM measurements showed that there was no surface morphology change during ageing for PEDOT:1.6PSS, PEDOT:2.5PSS and sheared films. 1.6PSS and 2.5PSS films showed increase in phase angle after ageing which could be attributed to the phase separation of PEDOT and PSS resulting in higher heterogeneity. DMSO/EG added films on the other hand showed no change in surface morphology or phase angle, which can be attributed to the breaking-up of large, ordered aggregates to smaller aggregates during ageing. PEDOT:1.6PSS, PEDOT:1.6PSS + DMSO and PEDOT:1.6PSS + EG showed increased rate of ageing in the presence of humidity. However, PEDOT:PSS with higher PSS content (2.5 PSS) and solution sheared films showed very little change in conductivity with ageing under humidity.

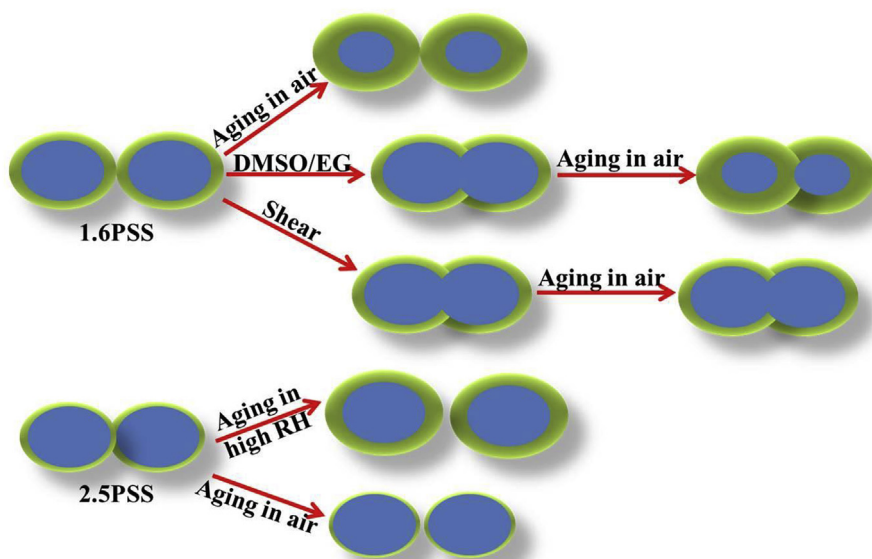


Fig. 13. Schematic illustration showing morphological changes during thermal ageing in PEDOT:PSS films in the presence and absence of humidity.

Author contributions

The manuscript was written with equal contributions from all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

Acknowledgments

FTIR and Raman Spectroscopy facilities were provided by SAIF, IIT Madras, India.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.polymdegradstab.2019.109025>.

ABBREVIATIONS

ATR-FTIR	Attenuated Total Reflectance- Fourier Transform
	Infrared spectroscopy
DMSO	Dimethyl sulfoxide
EG	Ethylene glycol

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